Table 1-1. *Abbreviated Hardware List*

Part	Description	Price (US / EUR)
Arduino Uno	Microcontroller interface board	$30/€22
USB cable	A-B (often used for printers) for the Arduino Uno	$4/€2
Solderless breadboard	Either standard (400 tie points) or mini (170 tie points)	$6/€4
Hookup wire	Solid 22 AWG or pre-terminated M/M jumper wires	$7/€5
Light emitting diodes	Standard 5mm LEDs and common cathode RGB LEDs	$10/€7
Resistors	Handful each of 220, 330, 1K, 10K ohm ¼ watt or similar	$10/€7
Capacitors	A couple 1 and .1 microfarad capacitors of any variety	$4/€3
Analog sensors	10K ohm trimpot, 10K ohm photocell, TMP36 temp sensor	$4/€3
Switches	Miniature momentary pushbuttons and tilt switches	$3/€2
Transistors	2N2222 or 2N3904 NPN transistor or similar	$1/€1

This minimal list of hardware will get you through about half of the book with other specific components necessary for the more advanced projects. When required, information on additional parts will be provided at the start of the chapter or section. While detailed in each chapter, these projects will include the use of temperature sensors, force/flex sensors, light sensors, accelerometers, wind sensors, liquid crystal displays, stepper motors, hobby servos, and other sorts of hardware. Although, rather than mandate a hefty toolbox of equipment, the projects and code will be written with as much flexibility as possible to allow the use of other sensors or actuators, depending on what you have available and your own personal interests and project ideas.

To keep things simple, we will power the Arduino Uno from the USB port of our computer using a USB cable for each one of the projects in this book. Once we upload the programs to our boards, however, the program is permanently stored on the board just like a USB flash drive. We could (if we wanted) power the Arduino board from an external power source, such as DC power supply or battery, and not need the USB cable at all. For more information, refer to the section "Connecting & Powering Arduino" on the Arduino Playground at `http://arduino.cc/playground/Main/ArduinoCoreHardware`.